Errata to "Dimension of bi-degree (d, d) spline spaces with the highest order of smoothness over hierarchical T-meshes"

The article [1] contains a few local counting and notational errors. The principal corrections are listed below; the proof structure is retained. A revised manuscript will be made available as arXiv:2507.11047v3, with all changes relative to the published journal version highlighted in blue. Locations below refer to the journal version.

1. **Pages 2203–2207, Theorem 3.4 and its proof.** In the diagonalizable case, the dimension formula should read

$$\dim \mathsf{CVS}[L(\mathscr{T})] = m(p-d-1) + n(q-d-1) - mn.$$

The term $-mn$ removes the double counting of horizontal–vertical intersections. The non-diagonalizable formula $(p-d-1)(q-d-1)$ is unchanged. Coordinate indices, block sizes and summation ranges in the matrix proof are corrected in v3, without changing the elimination sequence.

2. **Pages 2210–2212, Theorem 4.2 and its proof.** The corrected equation (4.1) is

$$\dim S_d(\mathscr{T}) = v + db + (d+1)^2 + \gamma_0 - (d+1)t. \tag{4.1}$$

Here v counts distinct vertices, t counts maximal l-edges including the four boundary l-edges, and b counts distinct boundary vertices including the four corners. The definition of γ_0 is unchanged.

The boundary-extension count on page 2212 should be

$$\bar{v} = v + 4d^2 + db + 4d, \qquad \bar{t} = t + 4d.$$

Indeed, the added vertices number $d(b-4) + 8d + 4d^2$: each original corner contributes in two directions. The omitted $4d$ changes the constant in (4.1) from $(d-1)^2$ to $(d+1)^2$. The level-counting notation in the proof is clarified in v3.

3. **Pages 2213–2216, Section 5 (the CVR graph).** The spaces compared are $\bar{S}_d(\mathscr{T})$ and $\bar{S}_{d-2}(\mathscr{C})$. In Lemma 5.3, write $\mathsf{CVS}_2[L(\mathscr{C})]$, not $\mathsf{CVS}_2[L(\mathscr{T})]$. In Theorem 5.5, the original-mesh contribution is $\sum_{i=1}^{\mathrm{lev}(\mathscr{T})} \dim \mathsf{CVS}_1[\bar{T}_i]$, not a sum with CVS_2. If m, n count the horizontal and vertical grid lines of $\mathscr{T}_0$, the initial dimensions are

$$\dim \bar{S}_d(\mathscr{T}_0) = \dim \bar{S}_{d-2}(\mathscr{C}_0) = (m-d-1)_+(n-d-1)_+,$$

where $(u)_+ = \max\{u, 0\}$.

In Lemma 5.4, construct C_2' by subdividing all $4(d-1)^2$ level-1 cells in the original $(d-1)\times(d-1)$ parent region and then taking the level-2 part of its CVR graph. The subsequent counts and the dimension equality stated in Theorem 5.5 are unchanged. Other degree, subscript and reference corrections are recorded in v3.

Clarifications of the preliminary definitions, notation and intermediate counts, including the auxiliary mesh in Theorem 3.8, and minor reference corrections are incorporated in v3; details are not repeated here.

References

[1] B. Huang, F. Chen, Dimension of bi-degree (d, d) spline spaces with the highest order of smoothness over hierarchical T-meshes, *Science China Mathematics* **69** (2026), no. 8, 2195–2218. doi:10.1007/s11425-023-2499-9.